The BrAIn Processing Unit: A Self-Assembling Stochastic Substrate for Brain-Like and Frontier Models

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Abstract

We propose the BrAIn Processing Unit (BPU), a non-von Neumann computing architecture whose primitives are brain circuits and whose models assemble themselves in hardware. Synapses are stochastic MoS_2 memristors that store and compute in the same device; a fire-together-wire-together substrate grows, strengthens, and prunes connections at runtime; and device variability is budgeted to a 2–6% band in which it regularizes learning, powers a local gradient estimate, breaks symmetry between duplicated modules, and frees attractors from mismatch pinning. Continuous attractors serve as registers, grid codes as addresses, scaffold memory as storage, and runtime growth as compilation. Because modern Hopfield recall is mathematically identical to transformer attention, the same substrate runs brain-like cognition natively and frontier models through a thin compatibility layer. We formalize the architecture as a dynamical system on five separated timescales, show that its averaged dynamics perform noisy gradient descent on a flatness-regularized objective combined with noise-guided search over the model's own structure, and define a verification method based on persistent homology that certifies the topology of structures the hardware grows. We report a first simulation campaign against pre-registered success criteria: attention through noisy crossbars recalls 32/32 patterns at every noise level tested, a ring attractor self-organizes from sparse random connectivity and certifies topologically in two of three seeds, and a local grow rule matches gradient-based sparse training; the predicted attractor noise optimum and modular self-assembly did not reach significance at the scale tested. A four-tile MoS_2 prototype (BPU-1) and an FPGA emulator sharing one device model follow.

1 Introduction

Every mainstream computer since 1945 separates a processor from its memory and executes a fixed program by moving data to instructions. In machine-learning workloads this transfer dominates energy: a 32-bit DRAM read costs roughly 640 pJ against roughly 3.7 pJ for a floating-point multiply [Horowitz, 2014]. Analog in-memory accelerators keep weights as conductances in crossbar arrays and remove the transfer, but they retain two assumptions of the conventional model: the structure of the computation is fixed when the device is programmed, and variability among analog devices is an error to be suppressed. Neuromorphic processors implement spiking neurons with fixed or externally programmed connectivity and do not run contemporary large models.

The BPU inverts all three choices. Its memory is its processor, its program is a shape the hardware grows into, and its building blocks are the circuits the brain uses to hold state, address memory, and reason. Table 1 states the correspondence.

Our contributions are: (1) an architecture in which brain circuits are the computing primitives and the model is a module graph the hardware grows (Sections 3, 5); (2) a local, noise-powered structural

Table 1: Each part of a von Neumann computer and its BPU counterpart.

Von Neumann computer	BPU
Processor and separate memory joined by a bus	Synapses that store and compute in the same device
Registers holding state	Continuous attractors holding persistent activity
Memory addresses	Grid codes: multi-module phase vectors
Reading and writing memory	Scaffold memory completed from partial cues
Search and lookup	Attractor recall, identical to attention
Program counter stepping through instructions	Path integration through a learned map
Compiler fixed at design time	Growth engine rewiring hardware at runtime
Noise treated as error	Noise budgeted as a resource

learning rule in which synapses and module connections grow where a node-perturbation estimate of the gradient is large (Section 5.2); (3) a unified account of noise as a resource with four roles that each predict a performance peak at moderate, nonzero noise (Section 6); (4) a topological verification method for self-assembling hardware (Section 7); (5) a device-to-circuit physical design on MoS_2, including the mechanism by which stochasticity enters and by which weights adapt (Section 4); and (6) a pre-registered simulation campaign with honest first results (Section 8).

2 Related work

Continuous attractor networks model persistent activity and path integration [Burak and Fiete, 2009, 2012]; grid cells form an exponential-range error-correcting code [Fiete et al., 2008, Sreenivasan and Fiete, 2011] whose toroidal population topology was confirmed with persistent homology [Gardner et al., 2022]; and grid attractors self-organize through spike-timing-dependent plasticity during exploration [Widloski and Fiete, 2014]. Vector-HaSH attaches content to a fixed grid scaffold for high-capacity memory without catastrophic forgetting; the Tolman–Eichenbaum Machine factors observations into structure and content [Whittington et al., 2020]; and grid-like codes index abstract conceptual spaces [Constantinescu et al., 2016]. Modern Hopfield networks recall patterns by an update identical to transformer attention [Ramsauer et al., 2021]. Three-factor rules link local plasticity to global objectives [Frémaux and Gerstner, 2016]; multiplicative STDP stabilizes weights [van Rossum et al., 2000]; homeostatic scaling rescues working memory under heterogeneity [Renart et al., 2003]; and cascades of synaptic variables give power-law forgetting [Benna and Fusi, 2016]. Sparse training grows connections by gradient magnitude [Evci et al., 2020, Mocanu et al., 2018]. Memristor arrays have hosted discrete and continuous attractors from precomputed weights, where read noise shifts the activity bump and write noise splits it [Wang et al., 2020]; oxide RRAM supports unsupervised STDP self-organization; and gate-tunable MoS_2 memtransistors are established [Sangwan et al., 2018]. Our earlier position paper found that spiking networks peak in accuracy at 2–6% hardware noise while convolutional networks degrade monotonically [Paradkar et al., 2026], and WHITEBOX introduced the topological metrics used here for monitoring [Paradkar and Paradkar, 2026]. No prior system grows attractors in hardware, uses noise as a structural signal, runs attention on the same substrate as brain circuits, or certifies what it grew.

2.1 TMDCs, MoS_2, and why variability has been treated as a flaw

Transition-metal dichalcogenides (TMDCs) are layered crystals of formula MX_2, a plane of a transition metal (Mo, W) between two planes of a chalcogen (S, Se, Te). Each three-atom sandwich is held to the next by van der Waals forces, so the material can be thinned to a single sandwich 0.65 nm thick, and a single layer of MoS_2 is a direct-gap semiconductor (about 1.8–1.9 eV) that can be gated, patterned, and transferred onto finished silicon at low temperature. The same thinness makes MoS_2 a natural memory element: a few sulfur vacancies or a few hundred metal ions change the conductance of the whole channel, so a device at the atomic limit stores an analog weight in a footprint set only by lithography.

That sensitivity is also why TMDC devices vary. Vacancy density, grain boundaries, and layer count differ from device to device; ion motion differs from pulse to pulse; and the conductance fluctuates between reads. Every prior line of work has treated this as the obstacle to deployment. Digital design hides it behind margins and error correction. Analog in-memory computing calibrates it out with write-verify loops, redundant devices, and higher-precision converters, and reports yield and mismatch as the main costs of scaling. The neuromorphic community characterizes it as mismatch to be compensated by on-chip learning or trimming. In each case the implicit goal is a device that behaves like an ideal number, and the price is paid in area, energy, and the exclusion of the cheapest and thinnest materials.

The BPU takes the opposite position, and it is the position the brain takes. Biological synapses are unreliable to a degree no engineer would accept (release probabilities of a few tenths, weights that drift over hours), yet the brain computes with them because its algorithms are built for stochastic parts: noise regularizes learning, provides the exploration that local credit assignment needs, breaks the symmetry between developing circuits, and keeps attractor states from freezing. Section 6 derives each of these roles for the BPU and shows that each predicts a performance peak at moderate, nonzero noise, which our position paper measured at 2–6% for spiking networks [Paradkar et al., 2026]. The engineering consequence is a reversal of the usual design goal: rather than suppressing variability, the fabrication step that sets the vacancy density is used to place σ_{eff} inside that band, and the architecture is designed so that the very vacancies that store the weight are the source of the fluctuation that the learning rules consume (Section 4).

3 Architecture

The BPU is a single dynamical system on five separated timescales. Its state is the triple (v, G, A): neuron activity, conductances, and a binary connectivity mask.

$$\tau_v \dot{v} = F(v,\, A \odot G,\, x), \qquad \dot{G} = \eta(V_G)\, \Phi(\text{pre}, \text{post}, m) + \xi(t), \qquad A_{t+1} = \mathcal{S}(A_t, G_t, C_t, \varepsilon_t), \tag{1}$$

with $\tau_v \ll \tau_G \ll \tau_A$ (milliseconds; seconds to minutes; minutes to hours). Each slower layer sees the faster ones as averaged out, which makes every layer analyzable on its own. Figure 1 shows the ten layers.

Physically, tiles ($n \times n$ crossbars, $n = 32$ on the prototype) form cores, cores form chips, and chips communicate through rewritable address-event routing tables. Inside a tile values stay analog; between tiles they travel as spikes or quantized vectors. Two operating modes share the fabric: *deploy mode*, in which a compiler maps an existing model's linear layers and attention heads onto tiles and programs conductances by write-verify, and *native mode*, in which the chip grows structure from a developmental seed under local rules.

4 Device physics and the origin of stochasticity

4.1 The synapse

A synapse is a channel of one to three layers of MoS_2 between electrodes (Figure 2). Monolayer MoS_2 is a sheet of molybdenum atoms between two sheets of sulfur, 0.65 nm thick, with a direct band gap near 1.8–1.9 eV. In the memtransistor form a gate sets the channel conductance by its overdrive, and memory enters as a threshold shift carried by an internal state $w \in [0, 1]$:

$$G = \mu C_{ox} \tfrac{W}{L} \big(V_G - V_{th}(w)\big), \qquad V_{th}(w) = V_{th,0} - \kappa w \;\Rightarrow\; G(w) = G_{\min} + (G_{\max} - G_{\min})\, w. \tag{2}$$

The gate scales the whole range without changing w; this is the terminal through which the neuromodulatory signal of Section 5.1 enters, and V_G at which plasticity vanishes is a physical kill switch for learning.

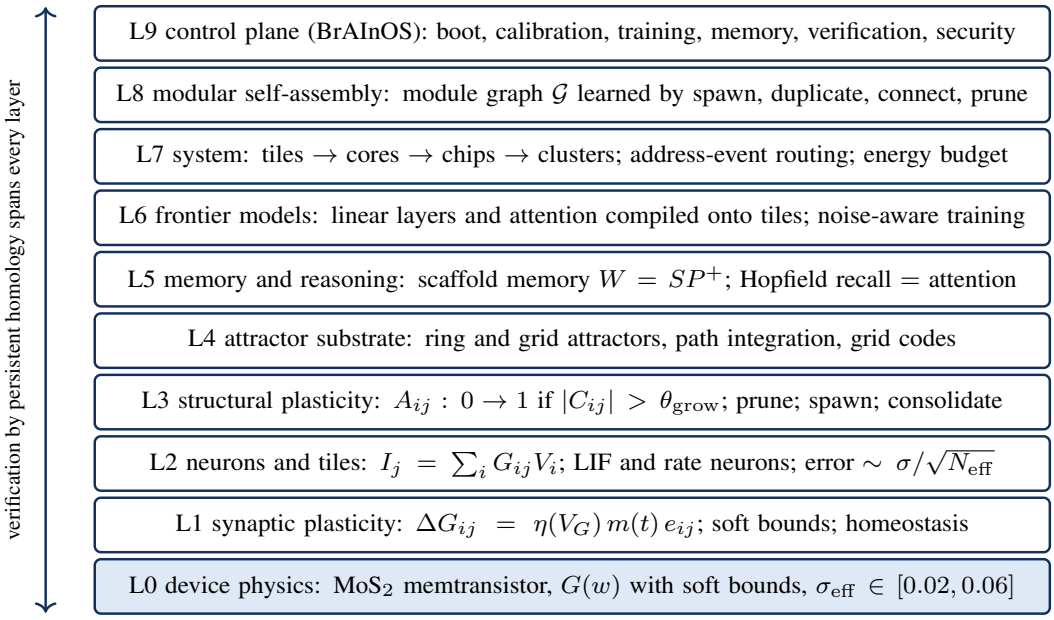


Figure 1: The BPU stack. The fabric (L0) is a uniform array of memristor tiles with no assigned function at manufacture; a thin fixed logic layer implements L1–L3 and L7, and the control plane (L9) coordinates the rest. A von Neumann chip is mostly logic with a little memory; a BPU is mostly adaptive memory with a little fixed logic.

4.2 How weights change physically

Memory is carried by the migration of sulfur vacancies, the growth of silver filaments along vacancy chains, trapped charge near the channel, or a local 2H→1T′ phase change (Figure 3). A potentiating pulse drives Ag^+ ions from the active electrode into vacancy sites; the filament lengthens and w rises. A depressing pulse of opposite polarity dissolves it. Each pulse moves fewer ions as the gap closes, because fewer mobile sites remain, which gives the soft-bound update

$$\Delta w^+ = \alpha_p e^{-\beta_p w}, \qquad \Delta w^- = -\alpha_d e^{-\beta_d (1-w)}, \qquad w(n) = \tfrac{1}{\beta_p} \ln(1 + \alpha_p \beta_p n). \tag{3}$$

With $\alpha_p = 0.05$ and $\beta_p = 3$, $w = 0.5$ takes about 23 pulses and $w = 0.9$ about 93: the device slows as it fills instead of hitting a wall, which Section 5.1 shows is exactly the stabilizer Hebbian learning needs.

4.3 Where the noise comes from

Three independent relative perturbations enter the device, each with a distinct physical cause (Figure 3c):

$$\alpha_{ij} = \alpha(1+\sigma_{d2d}\zeta_{ij}), \quad \Delta w_{ij} \leftarrow \Delta w_{ij}(1+\sigma_{c2c}\epsilon_{ij}(t)), \quad G_{ij}^{\text{read}} = G_{ij}(1+\sigma_r \nu_{ij}(t)), \quad \sigma_{\text{eff}} = \sqrt{\sigma_{d2d}^2 + \sigma_{c2c}^2 + \sigma_r^2}, \tag{4}$$

with ζ, ϵ, ν standard normal. Device-to-device variation (ζ, drawn once) comes from the random vacancy density, grain boundaries, and film thickness of each device. Cycle-to-cycle variation (ϵ, per write) arises because each pulse routes ions along a different vacancy path, so filament geometry and the step size differ from pulse to pulse. Read noise (ν, per read) is the thermal and random-telegraph fluctuation of the filament gap; it changes what the circuit sees, not what is stored. The design target is $\sigma_{\text{eff}} \in [0.02, 0.06]$, set at fabrication by the anneal or plasma step that fixes the vacancy density and, in operation, by injecting peripheral fluctuation when measured noise is below the band. Retention follows $G(t) = G_\infty + (G_0 - G_\infty)e^{-t/\tau_{\text{ret}}}$; diffusive silver filaments (τ_{ret} of ms to s) serve as working memory and attention cache, and vacancy or phase-change switching (years) holds long-term weights.

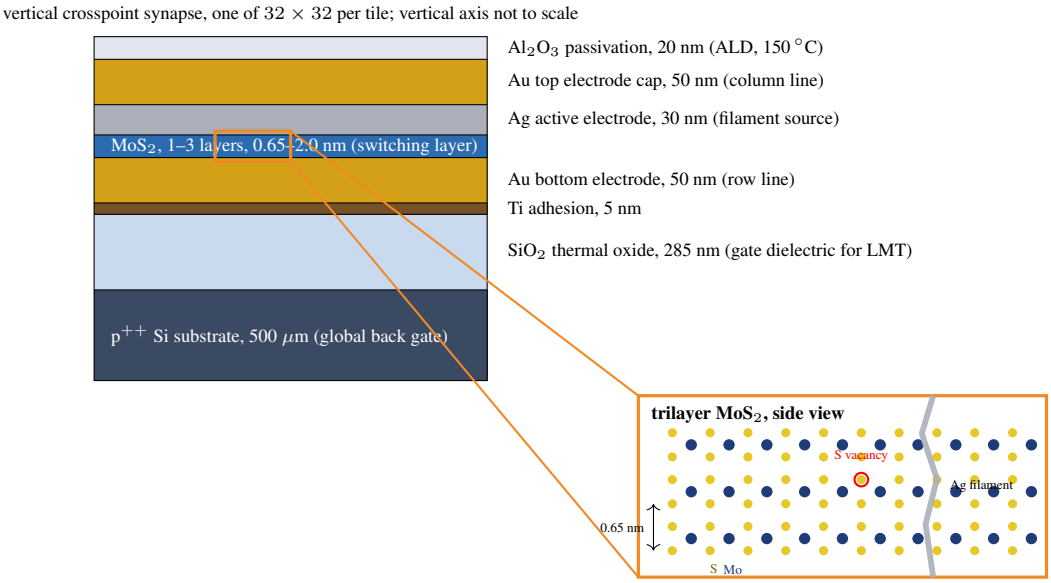


Figure 2: Layer stack of one vertical crosspoint synapse with thicknesses, and (inset) the trilayer MoS_2 switching layer at atomic resolution: sulfur–molybdenum–sulfur planes 0.65 nm apart, a sulfur vacancy, and a silver filament threading vacancy sites between the electrodes.

4.4 Tile, periphery, and die

Driving the rows of a tile with voltages V_i and holding the columns at virtual ground produces column currents $I_j = \sum_i G_{ij} V_i$, a complete matrix-vector product in one read; signed weights use differential column pairs, $W_{ij} = (G^+_{ij} - G^-_{ij})/G_{\text{scale}}$. With independent relative noise σ per device, an output has mean equal to the ideal product and relative error $\sim \sigma/\sqrt{N_{\text{eff}}}$, so a 4-bit device feeding a 256-input column yields 0.25% output error. Writes use a $V/3$ scheme (selected row $+V_w$, selected column 0, other rows $V_w/3$, other columns $2V_w/3$) so half-selected devices stay below threshold. The fixed logic at the tile edges (Figure 4) comprises transimpedance amplifiers and leaky integrate-and-fire neurons on the columns, STDP pulse shapers on every line, a write-verify controller per tile, and an address-event router with permission-controlled tables. The prototype die (Figure 5) carries four 32×32 tiles, one 64×64 array, and test structures on a four-mask process with no ion implantation and no step above 300 °C after the MoS_2 transfer.

4.5 BPU-3: the monolithic chip

BPU-1 is a hybrid: its synapses, plasticity, noise, and physical connectivity are real MoS_2, while its neurons, growth engine, converters, and router live on an FPGA and an interface board. BPU-3 moves all of that into a CMOS base layer and stacks the fabric above it (Figure 6). Each core carries a small RISC-V control core running BrAInOS, 128 neurons, the converter chain, the event router with its permission bits, the write-verify state machines, and the security root (secure boot and a physically unclonable function derived from the fabric's own variability). Four MoS_2 tile layers sit above the metal stack, each reached through vertical vias, with adjacent layers sharing an electrode plane. Two dedicated rails run to every tile's local gate: the modulator $m(t)$, which makes the three-factor rule act through device physics rather than pulse shaping, and the plasticity voltage V_G, which freezes learning in hardware when set to zero. Because two-dimensional materials process below 300 °C, the stack is built on finished CMOS, which is the density path to frontier scale: at 100 nm pitch, four layers give 4×10^{10} devices per square centimetre, and a 70-billion-parameter model's 1.4×10^{11} differential-pair devices fit in a few square centimetres. The device model, tile arithmetic, routing primitives, and operating system are unchanged from BPU-0 and BPU-1, which is why results on the emulator and the prototype carry forward.

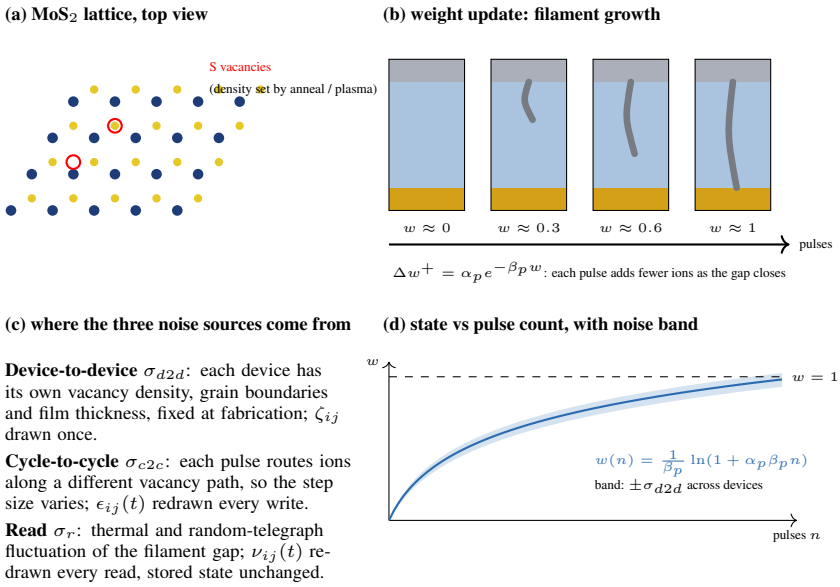
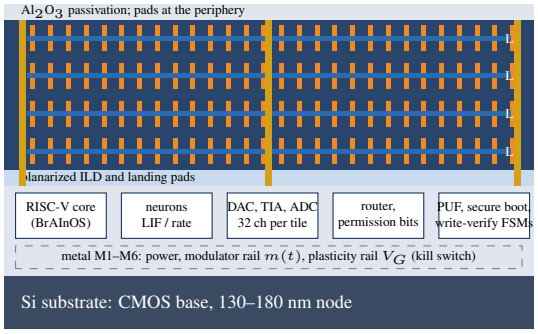


Figure 3: Molecular origin of adaptation and stochasticity. (a) Sulfur vacancies in the MoS_2 lattice are the mobile sites; their density is set by defect engineering. (b) A silver filament grows through vacancy chains pulse by pulse, raising w; growth slows as the gap closes. (c) The three noise sources map onto distinct physical causes. (d) The resulting state trajectory and the device-to-device band.



Four MoS_2 tile layers (L1–L4) above finished CMOS. Gold columns are through-stack vias; adjacent layers share an electrode plane, so four layers cost five electrode levels. Every MoS_2 step stays below 300 °C. Per core: 64 tiles, 65,536 synapses, 128 neurons, one control core. Per chip (8×8 cores): 4.2 M synapses at 2 μm pitch; at 100 nm pitch, 4×10^{10} devices per cm^2.

Figure 6: BPU-3 cross-section through one core: a CMOS base with control, neurons, converters, router, and security root; four stacked MoS_2 tile layers with through-stack vias; dedicated modulator and plasticity rails reaching every tile.

5 Mathematics

5.1 Synaptic plasticity

Pre- and post-synaptic neurons emit shaped waveforms; only their overlap exceeds the switching threshold V_{sw}, and the supra-threshold area reproduces the STDP window. STDP events charge an eligibility trace, and the modulator $m(t)$ converts the trace to a write:

$$\tau_e \dot{e}_{ij} = -e_{ij} + \tau_e \,\mathrm{STDP}_{ij}(t), \qquad \Delta G_{ij} = \eta(V_G)\, m(t)\, e_{ij}, \qquad \mathbb{E}[m\, e_{ij}] \approx -\partial \mathcal{L}/\partial G_{ij}, \quad (5)$$

the last relation being the policy-gradient identity of three-factor learning [Frémaux and Gerstner, 2016]. Pure Hebbian learning diverges along the top eigenvector of the input covariance; three brakes apply. The soft bounds of Eq. (3) make plasticity multiplicative, which yields a stable unimodal weight distribution [van Rossum et al., 2000]; Oja normalization converges to the unit-norm principal component; and homeostatic scaling $\tau_H \dot{G}_{ij} = G_{ij}(r^* - r_j)/r^*$ holds each neuron near a target rate.

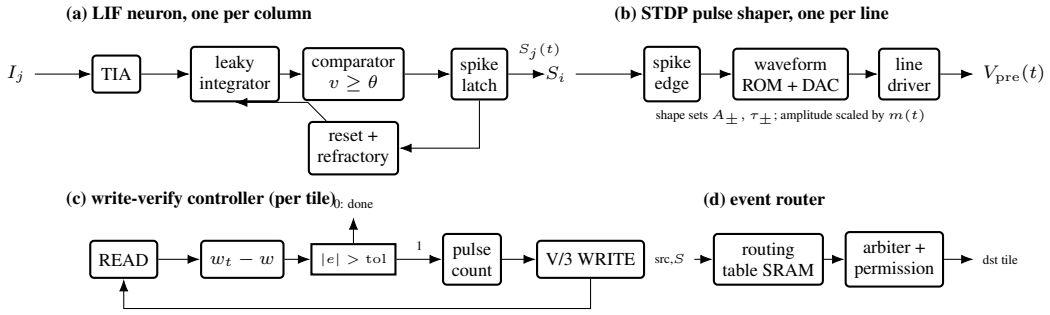
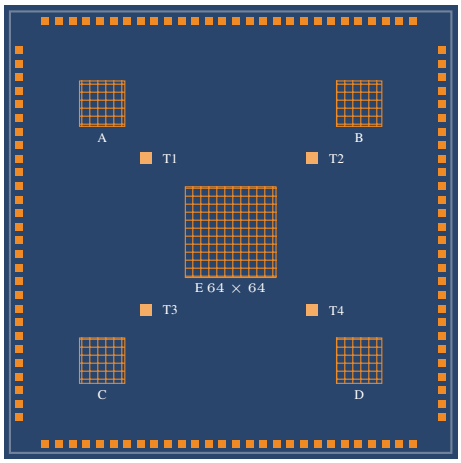


Figure 4: Fixed peripheral logic (gate-level sketch). (a) Column neuron: transimpedance amplifier, leaky integrator, threshold comparator, reset and refractory logic, spike latch. (b) Line driver: spike edge triggers a stored waveform whose shape sets the STDP window and whose amplitude is scaled by the modulator $m(t)$. (c) Write-verify controller. (d) Event router with routing-table SRAM and permission arbiter.



BPU-1 die, 10 × 10 mm. Tiles A–D: four 32 × 32 crosspoint arrays (ring attractor, grid module, scaffold memory, attention). E: 64 × 64 scaling array. T1–T4: 8 × 8 sneak-path test arrays. Single devices and lateral memtransistors (not shown) sit between tiles. 384 perimeter bond pads at 90 μm pitch; 20 μm crosspoint pitch; four masks.

Figure 5: BPU-1 prototype die floorplan (schematic).

Writes occur only when $|\eta m e_{ij}| > \theta_w$; with 10^6 cycles over 5 years the budget is 6×10^{-3} writes per device per second.

5.2 Structural plasticity

Weights and wiring are learned together against $\min_{G,A} \mathcal{L}(A \odot G) + \lambda \|A\|_0$. The ℓ_0 term has no gradient, so A changes by discrete events. Let each output fluctuate, $y_j = \bar{y}_j + \xi_j$, $\mathrm{Var}\,\xi = \sigma_\xi^2$, and broadcast $m = -(\mathcal{L} - \bar{\mathcal{L}})$. Because the ξ_k are independent with zero mean,

$$C_{ij} \equiv \langle m\, \xi_j\, x_i \rangle = -\sigma_\xi^2 \frac{\partial \mathcal{L}}{\partial y_j}\, x_i = -\sigma_\xi^2 \frac{\partial \mathcal{L}}{\partial W_{ij}}, \qquad A_{ij} : 0 \to 1 \text{ if } |C_{ij}| > \theta_{\mathrm{grow}}, \tag{6}$$

a node-perturbation estimate of the gradient at a synapse that may not exist. The factor σ_ξ^2 means that without noise the estimate is zero: device noise is what lets the chip sense where to grow. The rule is the local, hardware form of RigL [Evci et al., 2020]. Pruning releases synapses whose conductance stays within θ_{prune} of the floor for a window; dead devices are pruned automatically. Neurogenesis copies an overloaded module into a free tile; two identical copies under competitive learning obey $\dot{d} = \lambda d$ for their difference, so the symmetric state is unstable but is left only if $d(0) \neq 0$, which device noise supplies: $t_{\mathrm{split}} \approx \lambda^{-1} \ln(d_{\mathrm{target}}/\sigma_d)$. Consolidation pairs a fast volatile device g_f with a slow nonvolatile g_s, $\tau_c \dot{g}_s = (g_f - g_s)\,\mathbb{1}[\text{replay}]$, and chains of such variables give power-law forgetting [Benna and Fusi, 2016].

5.3 Continuous attractors and grid codes

A ring with $W(\Delta) = J_0 + J_1 \cos \Delta$ forms a stable bump when $J_1 > 2$ (a cosine ripple feeds back with gain $J_1/2$) and $J_0 < 1$. Translation invariance makes the derivative of the bump an eigenvector with eigenvalue zero, so the bump sits anywhere on a continuous ring of fixed points and its position ϕ is a stored analog value. A velocity-gated asymmetric term $W_v = W + \gamma v W' \approx W(\Delta + \gamma v)$ moves the bump at $\dot{\phi} = cv$. Under device mismatch the programmed weights are $W(\theta_i - \theta_j)(1 + \sigma\zeta_{ij})$; projecting onto the zero mode gives

$$\dot{\phi} = -\partial_\phi U_\sigma(\phi) + cv + \sqrt{2D}\,\eta(t), \tag{7}$$

with a random pinning potential U_σ [Itskov et al., 2011] and diffusion $D \propto 1/N$ [Burak and Fiete, 2012]. Noise frees the bump over a barrier ΔU at the Kramers rate $\propto e^{-\Delta U/D}$ while diffusing it, so path-integration error $E(D) \approx c_1 e^{\Delta U/D} + 2DT$ has a minimum at some $D^* > 0$: the attractor noise optimum, tested in Section 8. A two-dimensional sheet with difference-of-Gaussians connectivity forms a hexagonal lattice by a Turing instability at wavenumber k^* with spacing $\lambda = 4\pi/(\sqrt{3}k^*)$ [Burak and Fiete, 2009]; M modules with coprime periods encode position uniquely over $\prod_m \lambda_m$, an exponential range for linear cost [Fiete et al., 2008], and decoding to the nearest valid phase combination corrects errors [Sreenivasan and Fiete, 2011].

5.4 Memory and attention

Scaffold memory attaches content s to grid-indexed states p through $W = SP^+$, reached by the local delta rule $\Delta W = \eta(s - Wp)p^\top$; recall passes a cue through the grid attractor for cleanup. Modern Hopfield networks store patterns as columns X under $E(\xi) = -\beta^{-1} \log \sum_\mu e^{\beta x_\mu^\top \xi} + \frac{1}{2}\xi^\top \xi$, whose gradient gives the update $\xi^{\text{new}} = X\,\mathrm{softmax}(\beta X^\top \xi)$ [Ramsauer et al., 2021]: with $\beta = 1/\sqrt{d}$ this is transformer attention exactly. In hardware $X^\top \xi$ is one crossbar read, the exponential is native to subthreshold transistors ($I_D = I_0 e^{V_{GS}/nV_T}$), normalization is a current divider, and $X \cdot \mathrm{softmax}(\cdot)$ is a second read. With concepts in a grid-coded map, relations are displacements and analogy is phase arithmetic $\phi_d = \phi_c + (\phi_b - \phi_a)$; the Tolman–Eichenbaum factorization $x_t = f(g_t, s_t)$, $g_{t+1} = T(g_t, a_t)$ lets a learned structure transfer to new content [Whittington et al., 2020].

5.5 Modular self-assembly

A module is $\mathcal{M}_k = (G_k, A_k, F_k)$ with dynamics type F_k; modules connect through routing tables R and projections B, $u_l = \sum_k R_{kl} B_{kl} z_k + x_l^{\text{ext}}$. Weights, masks, connections, and module count are learned against

$$\min_{\{G_k\},\{A_k\},R,K} \mathcal{L} + \lambda_s \sum_k \|A_k\|_0 + \lambda_e \|R\|_0 + \lambda_m K, \tag{8}$$

by spawn, duplicate, connect (Eq. (6) applied between modules), and prune. Self-assembly is measured by Newman modularity Q of the routing graph, per-module task information $I(Z_k; T)$, and compositional transfer against a fixed architecture.

5.6 The whole machine, averaged

Averaging the synaptic update over fast neural noise gives $\langle \dot{G} \rangle \approx -\eta_{\text{eff}} \nabla_G \mathcal{L}_{\text{eff}}$ with $\mathcal{L}_{\text{eff}} = \mathcal{L} + \frac{\sigma^2}{2} \sum_i W_i^2 \partial_{W_i}^2 \mathcal{L}$, and the structural events perform stochastic local search over the objective above. The BPU is a physical implementation of noisy gradient descent on a flatness-regularized loss, combined with noise-guided search over its own architecture, built from brain circuits as primitives.

6 Noise as a resource

Noise does four jobs (Table 2). In each, the benefit rises quickly from zero and saturates while the cost is second order near zero and grows without bound, so the net $J(\sigma) = B(\sigma) - C(\sigma)$ has a single

interior maximum $B'(\sigma^*) = C'(\sigma^*)$, $0 < \sigma^* < \infty$. Our position paper found this shape empirically at 2–6% for spiking networks [Paradkar et al., 2026]. The BPU does not tolerate imperfect devices; it requires their imperfection.

Table 2: The four roles of device noise.

Role	Layer	Mechanism	Without noise
Regularizer	L6	$\mathbb{E}_\epsilon[\mathcal{L}(W(1+\sigma\epsilon))] \approx \mathcal{L} + \frac{\sigma^2}{2}\sum W_i^2 \partial^2_{W_i}\mathcal{L}$	sharp minima, no transfer to hardware
Gradient sensor	L3, L8	$C_{ij} = -\sigma_\xi^2\, \partial\mathcal{L}/\partial W_{ij}$	estimate is zero
Symmetry breaker	L3	$t_{\text{split}} \approx \lambda^{-1} \ln(d_{\text{target}}/\sigma_d)$	copies never specialize
Escape mechanism	L4	Kramers rate $\propto e^{-\Delta U/D}$	bump pinned by mismatch

7 Verification

Population activity of N neurons over T steps is a cloud of T points in $\mathbb{R}^N$. Its Vietoris–Rips persistent homology yields features with birth b and death d; a ring certifies with Betti numbers $(1, 1, 0)$ and a torus with $(1, 2, 1)$ when exactly those features outlive the 99th percentile of the longest feature in a null obtained by circularly shifting each neuron's record, and no extra persistent β_0 component appears. Over learning we compute Structural Velocity $\text{SV} = W_p(D_t, D_{t+\Delta t})/\Delta t$, Topological Complexity $\text{TC} = \sum(d - b)$, and Feature Persistence Lifetime, which in WHITEBOX separated memorizing from generalizing regimes and predicted per-class accuracy [Paradkar and Paradkar, 2026]; here they act as the chip's runtime health monitor.

8 Experiments and first results

Seven pre-registered experiments test the design with explicit refuting outcomes; the simulator implements the equations above on 32×32 tiles and shares its device model with the FPGA emulator. Table 3 reports the first campaign (laptop CPU, 16 October 2026, 3 seeds unless noted). Three further experiments (E8–E10: a transformer compiled onto noisy tiles with an energy model, on-chip trainability under distribution shift against a frozen deployment, and continual learning with consolidation) are running at the time of writing.

Table 3: First simulation campaign against pre-registered criteria. Reported as measured.

Exp	Prediction	Measured	Status
E1	GPT-2 124M within 2% perplexity at $\sigma =$ 2–6% after noise-aware fine-tuning	pipeline built; smoke test passes; GPU run pending	pending
E2	Local grow rule within a few percent of RigL at 10% density	local $0.914 \pm .003$, RigL $0.923 \pm .001$, random $0.914 \pm .002$	met; no edge over random
E3	Path-integration error minimized at nonzero noise	$0.38 \to 0.33$ ($\sigma = 0.05$) $\to 0.41$ rad; 7/10 seeds better, $p = 0.34$	not significant at $N = 32$
E4	Ring self-organizes; one β_1 bar, no extra β_0	2/3 certified, 0/3 controls, 6/8 ring coverage	partial
E5	Grid module self-organizes into a torus	0/6; programmed reference also 0/6	certifier and sheet size inadequate
E6	$\geq 90\%$ recall from corrupted cues under noise	attention 32/32 at every $\sigma \leq 0.2$; pseudo-inverse scaffold ≤ 8 pairs under 50% erasure	met by attention
E7	Modularity rises; modules specialize; grown graph beats fixed on a composed task	$Q = 0.12$; routes grow toward outputs; learner too weak; gap $+0.01$	null

What held. Attention through noisy crossbars is exact at every noise level, the ring attractor grows from sparse random connectivity and passes topological certification in most seeds, the grow rule is competitive with gradient-based regrowth, and the device model, tile arithmetic, and certifier behave as derived.

What did not. The attractor noise optimum appears as a dip inside the error bars at $N = 32$ and $T = 3$ s, where diffusion ($D \propto 1/N$) dominates; it is being re-run at $N = 128$–256 with homeostatic compensation and a direct measurement of the pinning barrier. Modular self-assembly grows routes in the right direction but node perturbation alone is too weak to train the modules; the re-run uses layer-local losses inside modules and keeps node perturbation for structural decisions. The torus test failed on its own positive control, so it measured the certifier, not the chip; a 5×6 sheet cannot host a two-dimensional attractor, which reassigns tile B of the prototype. Three findings bear directly on hardware: a 32-neuron ring needs a nearly complete neighbour mask, so the prototype yield gate must be tightened; whole-tile rewrites cost about half a pulse of precision each; and a developmental phase at the intensity used writes $18\times$ over the endurance budget.

9 What this does to AI compute

Three multipliers compound on a BPU, and none of them is available to a von Neumann accelerator (Figure 7).

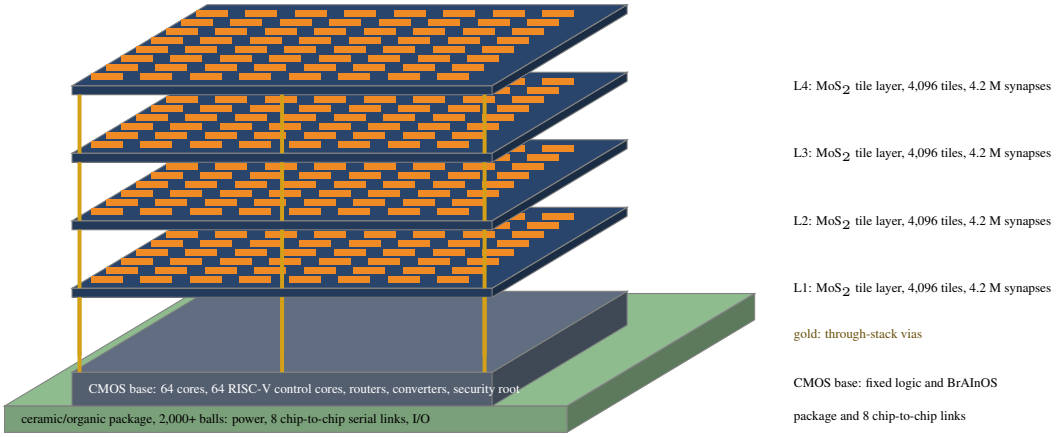
Energy: the weight never moves. Per token, a transformer performs about $2N_p$ multiply-accumulates for N_p parameters. On a GPU each costs a multiply plus, amortized over cache reuse, a share of a DRAM fetch; with the Horowitz constants (3.7 pJ per multiply, 640 pJ per 32-bit fetch) that is 644 pJ with no reuse and about 68 pJ with tenfold reuse. On a BPU the same operation is one analog read through a conductance, $E_{\mathrm{MAC}} \approx V^2 G t \approx 4$ fJ, plus conversion at tile edges, roughly one 8-bit conversion ($\sim$0.26 pJ) per thousand multiply-accumulates. Figure 8 plots both from 10^7 to 10^{11} parameters: the projected gap is about three orders of magnitude, and it is a projection from the device model with the assumptions stated, not a measurement. Even a projection that is wrong by $30\times$ leaves a $30\times$ advantage, because the dominant term, weight movement, is absent rather than reduced.

Density: the third dimension. Two-dimensional materials process below 300 °C, so tile layers stack above finished CMOS; four layers at 100 nm pitch give 4×10^{10} devices per square centimetre, and a 70-billion-parameter model fits in a few square centimetres of fabric with no off-chip weight memory at all. The frontier-model bottleneck shifts from memory bandwidth, which does not scale with transistor count, to device count, which does.

Learning that does not stop at deployment. A GPU-trained model is frozen when it ships; a BPU keeps learning in place with local rules under a write budget, consolidating what matters into nonvolatile devices during replay. Compute that improves during use is a different quantity from compute that is spent once, and it is the quantity that matters for agents that act in the world. The same mechanism makes the architecture itself a learned variable: experts, modules, and connections are recruited where the gradient estimate says they are needed, so the search over model structure that today costs thousands of GPU-hours per architecture happens inside the hardware as a side effect of running it.

Two models on one substrate. Because attention is Hopfield recall, the fabric that runs a frontier model natively also runs grid codes, attractors, and scaffold memory, which are the components of planning, episodic memory, and reasoning as navigation that transformers lack. A BPU is therefore

not only a cheaper place to run current models; it is the first place where the brain-circuit primitives and the frontier-model primitives can be composed in one learned module graph.



BPU-3, one chip (exploded view)
8×8 cores $\times$ 4 layers = 16,384 tiles
16.8 M synapses at 2 μm pitch; the same die at 100 nm pitch holds 4×10^{10} devices
weights never leave the fabric; only events and compressed representations cross the package
learns in place; freezes on a single rail ($V_G = 0$)

Figure 7: The final product: a BPU-3 chip in exploded view. A CMOS base carrying 64 cores of fixed logic and control; four MoS_2 tile layers above it joined by through-stack vias; a package that carries only power, events, and compressed representations.

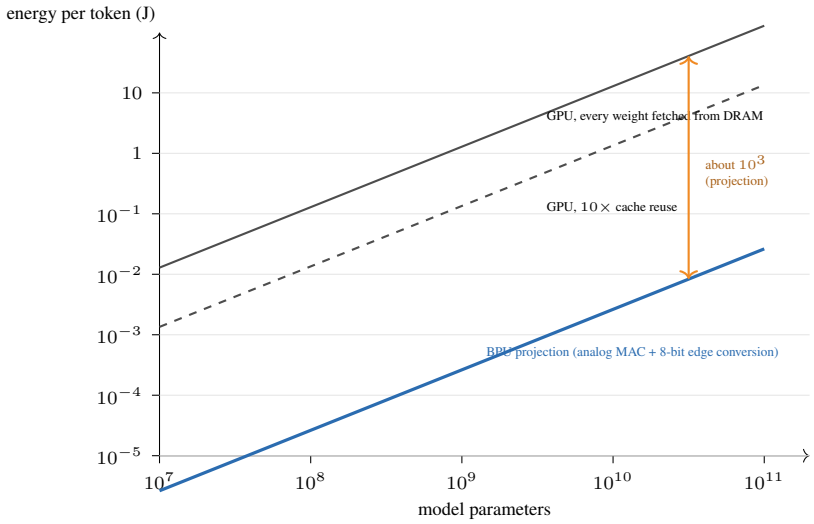


Figure 8: Projected energy per token against model size. GPU curves use literature per-operation constants [Horowitz, 2014] with no cache reuse and with tenfold reuse; the BPU curve uses the L7 device model ($V = 0.2$ V, $G = 10$ μS, $t = 10$ ns, 8-bit edge conversion at one per thousand operations). Projection, not measurement.

10 Limitations

BPU-1 is a hybrid with neurons and the growth engine on an FPGA, so system-level energy and latency are projections from the L7 model with literature constants. Vertical crosspoints have no

gate; the gate-borne modulator is demonstrated on lateral test devices until BPU-2 adds local gates. The noise optimum and self-assembly claims are hypotheses with refuting outcomes, and the first campaign did not confirm them. Credit assignment without backpropagation at frontier scale remains open.

11 Conclusion

The BPU proposes that the next computer should be grown rather than programmed: brain circuits as primitives, a fabric that stores and computes in the same device, noise budgeted as a resource, and topology as the audit. The first campaign establishes the substrate (attention, ring growth, local growth rules) and marks precisely where the stronger claims remain open. The FPGA emulator and the MoS_2 prototype will repeat every experiment on hardware.

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